

Project Design Phase Proposed Solution Template

Date	6 june 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Flooding causes massive loss of life, displacement, and infrastructure damage globally. Traditional flood forecasting models often rely solely on bulk historical rainfall averages, overlooking real-time microclimate indicators (temperature, humidity, cloud cover) and localized water basin indices. This leads to delayed, inaccurate, or non-actionable warnings. FloodSense AI solves this by predicting imminent flood risk using a combination of live atmospheric metrics and seasonal precipitation levels.
2.	Idea / Solution description	An end-to-end, intelligent flood risk assessment system. It consists of: 1. Machine Learning Engine: A pipeline (train.py) that trains and compares 5 different classifiers (Logistic Regression, Decision Trees, Random Forest, KNN, XGBoost), selecting the best-performing model (Random Forest, 96.55% accuracy). 2. Flask Backend API: A web server (app.py) that loads the model, standardizes user inputs, manages secure account credentials (hashed via Werkzeug), and records a persistent history of results using an SQLite database. 3. Interactive Dashboard: A glassmorphic dark-theme UI where users input 10 atmospheric and seasonal rainfall features to receive an animated probability gauge, risk classification, and contextual safety recommendations.
3.	Novelty / Uniqueness	• Multivariate Predictive Modeling: Blends standard precipitation values with real-time local microclimate indicators (Relative Humidity, Temperature, Cloud Cover) and a customized Sub-division Basin Runoff Index (sub). • Optimized Probability Gauge: Specifically leverages Random Forest to generate continuous, high-fidelity probability estimates

		(0-100%) rather than a simple binary (Yes/No) decision boundary. • Intelligent Safety UI: Directly translates numeric risk probability into real-time, actionable emergency directives.
4.	Social Impact / Customer Satisfaction	• Community Empowerment: Gives local residents and community organizers access to a localized warning tool, helping them prepare before official warnings are broadcast. • Mitigation Actionability: The UI displays customized safety plans based on danger levels (e.g., evacuation warnings vs. routine drainage maintenance). • Historical Audit Trail: Allows users to keep a profile-based history of entries to track changing environmental trends over time.
5.	Business Model (Revenue Model)	• B2B / B2G SaaS Subscriptions: Paid subscriptions for agricultural firms, insurance agencies, real estate developers, and local disaster response departments to access predictive APIs. • Freemium Tier: Free manual assessment tools for the public, with paid tiers for automated meteorological API integrations. • Consultancy Services: Custom training of localized sub-division models for municipal dams and hydroelectric agencies.
6.	Scalability of the Solution	• Decoupled Architecture: Separation of offline model training (train.py) and online inference (app.py) ensures millisecond-level response times. • API-First Framework: The prediction engine is served via a RESTful JSON endpoint (/predict), allowing the server to be scaled as containerized microservices (e.g., Docker, AWS ECS) or integrated into mobile apps. • Database Portability: The database schema uses standard SQL, making it easily migratable from SQLite to highly scalable cloud databases like PostgreSQL or MySQL.